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G4DARI user guide

This is a simplified free version of G4DARI.

GEANT4 Dose And Radiation Interaction (G4DARI) is a graphical interface designed to compute absorbed dose (Gy) and energy deposition (MeV) directly within a CT dataset expressed in Hounsfield Units (HU).

The platform replaces the complexity of traditional GEANT4 and GATE workflows by eliminating the need for command-line operations, scripting, or manual configuration of physics parameters. Instead, users conduct the full simulation process through structured menus and editable fields that capture all required inputs.

Installation procedure

1. Create a directory *C:\gate_dosimetry* and within it four subfolders named *apps*, *data*, *mac*, and *output*. These folders will respectively store installers, common files for all simulations and CT image files, GATE macro files, and simulation results.
2. Download Oracle VirtualBox (OVB) from the official website and save the installer in the *apps* folder. Install it using the default settings.
3. Download the file *vGate_9.4.1.ova* from the official GATE distribution website (<https://gate.uca.fr/download/vgate?utm#/admin>) and save it in *apps* folder.
4. In OVB, select File - Import Appliance (Fig. 1). Click the folder icon at the end of the input line and select *Apps/vGate_9.4.1.ova* to import the virtual machine.
5. VirtualBox will display the default appliance configuration, including CPU allocation, RAM, storage, and network settings. These presets are suitable for *vGate*, and they can be modified later from OVB – File – Preferences, from OVB - Settings or from VM – Machine - Settings.
6. In the OVB main window, select *vGate_9.4.1* and click the green arrow *Start*. VM opens and requests a username and password. Use "**vgate**" as the username and "**virtual**" as the password (Fig. 2).
7. To enable shared folders between Windows and VM, and to use other utilities, install the VirtualBox Guest Additions in VM. From VM menu, select Devices - Insert Guest Additions CD Image. Open a terminal in VM (the black icon in the column of icons on the left in VM), and type: `sudo apt update` (followed by enter key), then

`sudo apt install build-essential dkms linux-headers-$(uname -r)` (followed by enter key). There are many guides available online that can be consulted, and the best interactive guide is with ChatGPT.

8. After Guest Additions is installed, configure the shared folder. Quit VM. In OVB, with vGate_9.4.1 selected, choose Settings - Shared Folders, click the green plus sign, select *C:\gate_dosimetry*, and enable "Auto mount".
9. Download and install PuTTY from its official website. Close VM. In OVB, open Settings - Network for vGate_9.4.1 and set "Adapter 1" – "Attach to" as "NAT". Start VM again, open a terminal, and type "ip addr" to view the "inet" IP address (often 127.0.0.1 when port forwarding is used). Open PuTTY app from Windows Start menu, then typing PuTTY on top of the window, then configure Host name as 127.0.0.1, Port as 2222, Connection type as SSH, and use the credentials vgate and virtual.
10. In VM terminal from home directory, open the file .bashrc with an editor such as gedit (`gedit .bashrc&`), add in this file: `cd /media/sf_gate_dosimetry/mac`, and: `sim=Gate main.mac`. After saving the changes in .bashrc, in the terminal, make: `source .bashrc`. This way, in PuTTY, the default folder is *C:\gate_dosimetry\mac*, and to run a simulation, just type "sim". Since PuTTY is installed, modifying .bashrc can also be done from PuTTY. To open PuTTY: Windows Start – and look for PuTTY application. Later, after first installation, when G4DARI is correctly running, OVB, VM and PuTTY can be started from G4DARI main menu with a single click on "Start_PuTTY". The user doesn't interact with OVB and VM, they can configure the simulation in G4DARI, and only type "sim" in PuTTY, then after the simulation run, they can use G4DARI to display and work with their results.
11. Before using G4DARI, which is a deployed Matlab GUI G4DARI.exe, MATLAB_Runtime_R2022a_Update_9_win64.zip (3.7 Gb) has to be installed from <https://www.mathworks.com/products/compiler/matlab-runtime.html>.
12. To begin a simulation project, double-click G4DARI.exe in Windows *C:\gate_dosimetry\apps* to open the interface (Fig. 3 and 4). From G4DARI menu, select Start_PuTTY to sequentially open VirtualBox, Virtual Machine and PuTTY terminal.

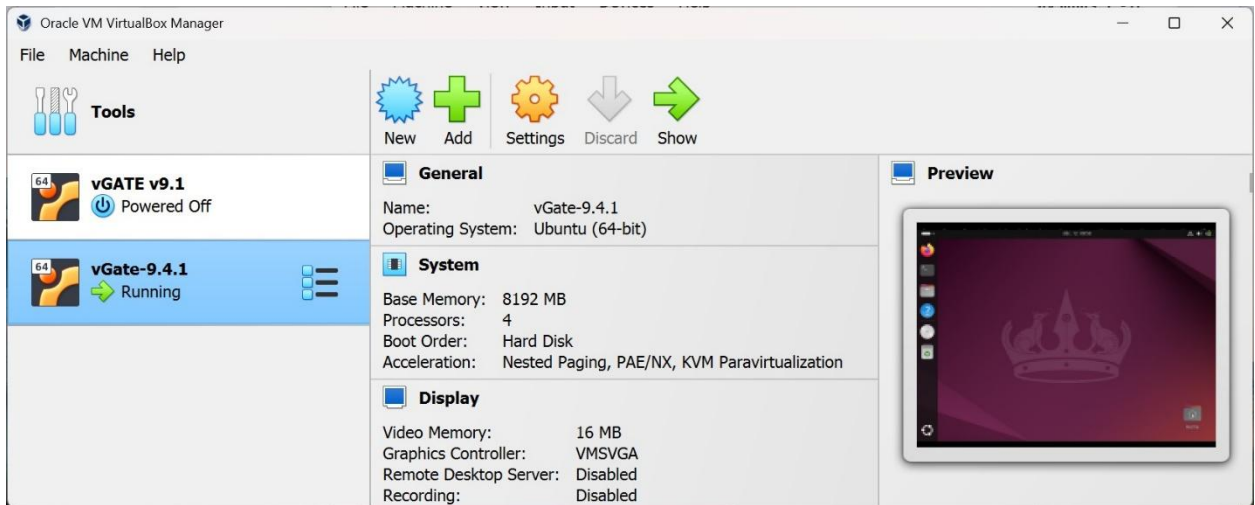


Fig. 1. Oracle VM VirtualBox with vGate_9.4.1 selected.

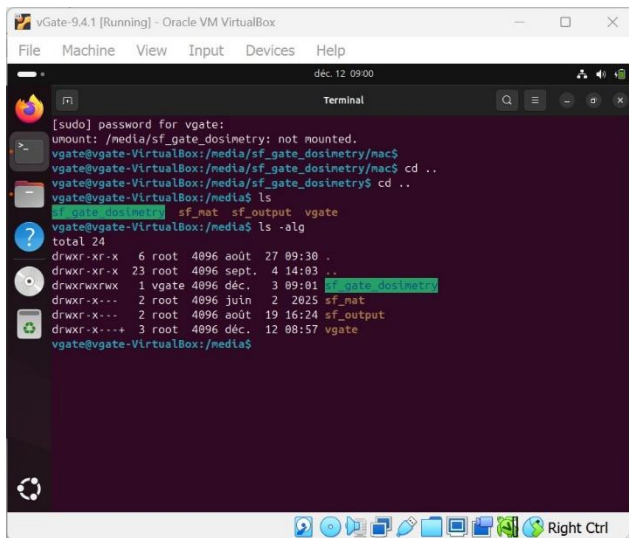


Fig. 2. vGate_9.4.1 Virtual Machine from which the simulation should be launched when using written simulation codes. With G4DARI, no interaction with this window is necessary. It is started by a click in G4DARI on Start_PuTTY.

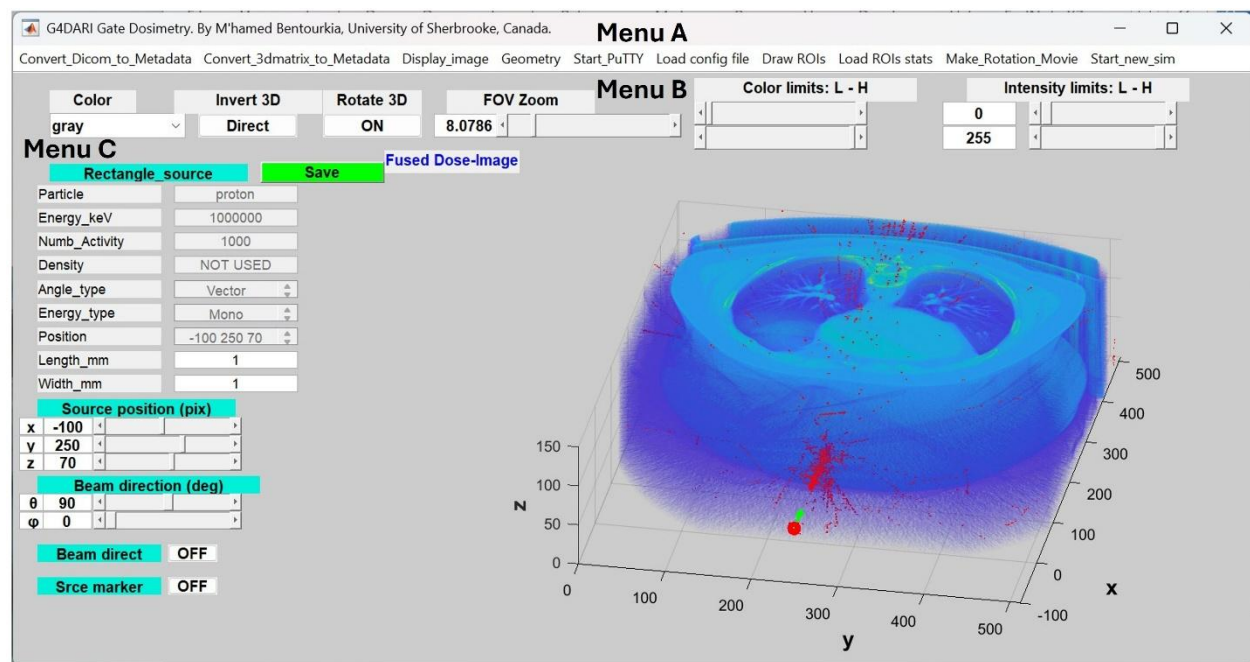


Fig. 3. Simulation setup with a thorax CT image. The menu on the left shows the parameters of the simulation. The red dot and green arrow help to visualize the source position and direction of emission. Here 10^3 protons of 1000 MeV were simulated.

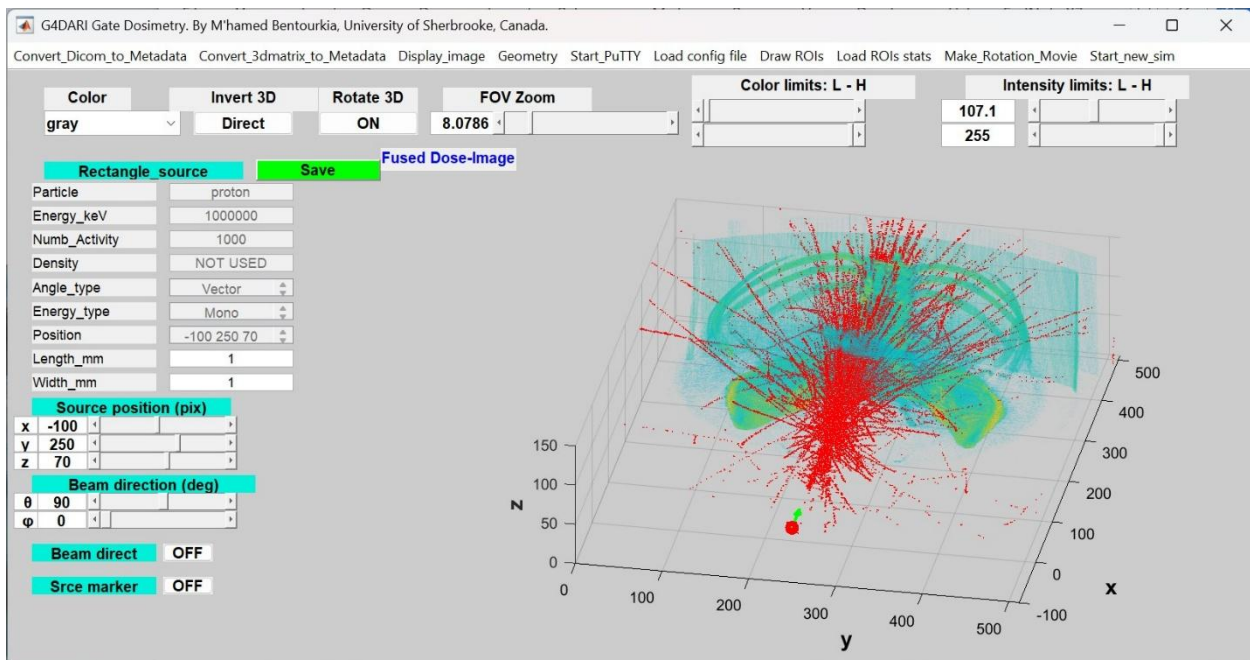


Fig. 4. Same setup as in Fig. 3 with thresholding (menu Intensity limits) of the CT image to see the protons inside the thorax.

Usage of G4DARI

Main menu (Menu A in Fig. 3):

1. Convert_Dicom_to_Metadata: select a folder containing the DICOM files to convert. Two files *.mhd and *.raw are written in *C:\gate_dosimetry\data* folder. These two files must have the same name and cannot be changed (unless you change the raw file name within mhd file as `mhd.ElementDataFile='filename.raw'`). All the files created are appended with the date and time suffix to differentiate them. This is helpful when repeating the simulation.
2. Convert_3dmatrix_to_Metadata: an mhd and raw set can be created given an image in Hounsfield units. The file name is appended with "Mx" for matrix data. The matrix has to be written in format *double* or *int16* in a file with *ieee-le* and extension *.dat*. If it is written with Matlab in a "mat" file, the matrix must be named "img". The user is asked to supply the image voxel size and dimensions. This is helpful to simulate any image as a phantom, not only DICOM images.
3. Display_image: a) Load_CT_image: select an mhd file from *data* folder to load the corresponding image in 3D, to zoom, rotate, threshold. b) Adjust_image: it happens the image has to be rotated or flipped. c) Save_adjusted_image. d) Load_dose_image: select the *Dose.mhd or *Edep.mhd from *output* folder to see the particles and their interactions. e) Locate_Dose: allows to superpose image of the dose or deposited energy on the CT image. Use intensity thresholding (Intensity limits) to better see the particle interaction positions.
4. Geometry: World_geometry: this is to start the configuration of the simulation. It opens the GATE "World" fields to enter the size of the space containing the image and the emitting source.
5. Start_PuTTY: when the simulation is configured (or before), click on this menu to open VB, VM and PuTTY where to type: "sim", to run the simulation after configuration.
6. Load config file: when the configuration is completed, or later after the simulation, this menu allows to open the simulation configuration. Again, all the files from a simulation have the same prefix and each file is suffixed with date and time. Another file is also useful for information and is generated at the end of the simulation: *SimStats.txt.
7. Draw ROIS: it opens a window where 4 views are ready to display the CT image to load on which to draw one or more ROIs, on one or more slices. The user finds information in each dialog. After deciding an ROI to draw, a "Zoom on" can be applied to the image in the fourth panel by dragging only once in the image, after that the "Zoom on" must be ticked off, and then start drawing the ROI by pressing the mouse and dragging it till the ROI is closed. The ROI is saved as a binary file *rois.mat. This *rois.mat can be applied on the dose or deposited energy image to estimate the absorbed dose or deposited energy at the ROI location. A file with same name and having extension *.txt provides the mean, standard deviation and maximum intensity of the ROI in that image.

8. Make_rotation_movie: for a presentation, showing the image rotating is more meaningful than a static image.
9. Start_new_sim: this is used when the CT image has to be changed and start a new simulation.

Tools menu (Menu B in Fig. 3):

Invert 3D: allows to flip up-down the image (head-first or feet-first).

Color limits and Intensity limits: they act on image thresholding.

Simulation configuration (Menu C in Fig. 3):

First, start the simulation by clicking on "Geometry" in the Menu A. This displays the x, y and z dimensions of the space (World) containing the simulation (source and image). Then click on "Save". This opens the window as in Fig. 3.

Just click on the space on the right of each item and accept or enter the values.

Some menus have submenus like:

Numb_Activity: select "Number" to enter a number of particles, or select "Activity" to enter the "Activity" and "Time Step", "Time Start" and "Time Stop".

Angle_type: select "Vector" to use a beam line, or "Angles" to use an opening angle from z-axis (theta) and x-axis (phi).

Energy-type: select "Mono" to use the energy defined above or "Spectrum", and in this case a dialog opens prompting to select a text file with two columns: energy and values of the spectrum.

Position: the position of the source. This can be defined here or in x, y and z under "Source position (pix)". The values of the source position cannot be out off the World. They can be within the image matrix or outside. The red dot and the green arrow help the user to locate the position of the source and the orientation of the beam. In case of angles of emission are selected in "Angle_type", the angles of emission or green arrow are placed at the mean value of the min and max theta and phi angles, and this is automatically updated in the editable fields "Beam direction (deg)".

Beam direction (deg): when "Angle_type" is not selected as "Angles", use here the beam direction theta and phi of the particle emission.

"Beam direct" and "Srce marker" On or OFF: allow to display the source position and direction of emission in the image. Removing them is useful to use the figure in a presentation or a document.

Running test simulations with supplied files:

All the necessary files are supplied in the folder data.

Clicking on "Geometry" menu then "World Geometry" displays the World dimension values, and after pressing "Save", asks for an image to load. Load "FP_ct11_abdomen_img_118sl_04Dec2025_15h15m28.mhd". Three other images are supplied in *data*: a thorax image with a 4 cm diameter spherical tumor located at [300,200,64]: FP_ct11_abdomen_img_118sl_04Dec2025_15h15m28_wtumor.mhd; two phantom images of water and bone (1000 HU). If needed, the image *mhd* files can be open with a text editor to read image information.

Once the image is displayed, set "Rotate 3D" to "On" to rotate the image to better locate the source position and direction of emission (red dot and green arrow). Use "FOV zoom" slider to adjust the image size till to see the axes x, y and z.

Fill all configuration fields, then press "Save". This creates the configuration file that you can display with "Load config file" for information or for reuse of simulation configuration. Press "Start_PuTTY", and once open type "sim".

After the end of the simulation, the output files *Dose.mhd and .raw are saved in the "output" folder.

To see the distribution of the dose, click on "Display_image" then "Load_Dose_image".

To superpose the dose map on the CT image, click on "Display_image" then "Locate_dose". In some cases, use *Edep.mhd which is energy deposit, better visualized than the dose.

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